Environment Similarity Index: Quantifying map-environment divergences for robot localization performance estimation and improvement

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***Abstract*— Should take about this much space: abc.**

I. Introduction

An indoor environment that often changes it's layout has long been a major challenge in terms of reliable localization for Autonomous Mobile Robots (AMR's) (ref). Being able to localize is a fundamental ability required to perform any kind of autonomous task (ref), and this can be greatly compromized when objects are no longer where the robot expects them to be (ref). Particle filters such as Adaptive Monte-Carlo Localization (AMCL) have long been the gold standard (ref) within the industry, due to its robustness to minor Map-Environment Differences (MED's) and sensor noise, and its low computational complexity. Although AMCL is reliable in many cases, it assumes the world is static, and does not update the map over time. If the MED becomes too great, it will eventually diverge and fail (ref). Other methods such as Simultaneous Localization and Mapping (SLAM) variations do have continuous mapping capabilities, but they too are succeptible to changes (ref). All it takes to corrupt the map is one wrong data association, and once enough errors have accumulated, the map becomes effectively useless, which is why SLAM suffers in long-term reliability. A possible solution is to modify the infrastructure by adding i.e. beacons (ref) or fiducial markers (ref), but this is not always an option; and even if it was, it may be too expensive and/or time consuming to be considered. Seeing as changes made to the environment compared to the robot's reference is a fundamental issue, it stands to reason that many AMR's would benefit greatly if they were able to detect and quantify this difference. However; to the best of our knowledge, no such metric exists. These are the contributions of this paper:

We propose the **Environment Similarity Index** (ESI): An index that quantifies how well the currently observed environment matches the map. First, we will use a simulation to show that the ESI is a reliable measure of the true MED, and assess their correlation. Then we will show that there is a clear correlation between the observed ESI score and the position RMSE of AMCL, compared to ground truth. In addition, we will show how the ESI can be used actively to to improve AMCL's performance by dynamically adjusting it's parameters. Finally, we will also demonstrate that the ESI can be calculated on real-world data in a real industrial facility, and show how it can be used to generate a detailed heatmap of the site that can be used for a multitude of purposes, such as identification of hard-to-localize areas, and detection and tracking of site activities. The real-world data will also be used to confirm that the ESI is truly a measure of the MED, and is not significantly affected by pose errors caused by changes in the environment, as long as these remain below a certain level.

II. Related Work

- Bayes filtering / Probabilistic map update - Scan-BIM divergences - Occupancy grid vs feature maps -

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III. Methodology

This section describes how changes in the environment are defined, and how they are used to calculate the ESI.

A. Changes

The difference that is measured is defined as a *change in how the robot perceives its environment*. The change is measured against the map the robot has of said environment. This is an important definition, with three major aspects. The first is that there can be a lot of changes to an environment, but if the robot does not sense it, then the changes by definition have no impact on the robot's ability to localize itself. The second is that it makes the index agnostic to which localization method is used, and which sensor modalities are available. A difference in observed visual features (I.e. SIFT) can be used just as well as geometric lidar, radar or sonar points, both 2D and 3D. The third is that changes that *could* be sensed but aren't, should not affect the index. Changes that happen outside the sensor coverage can naturally not be detected, and so they have no effect on localization. Unseen changes may also change back to the initial state before they are even noticed. The metric should as mentioned apply to any kind of sensors used, but since MiR robots localize via 2D Lidar, this will be the sensor modality used from here on. The reference map could likewise be of any kind (graph based, geometry based, grid based, etc), but for this project, the (occupancy) grid-based map is selected.

Any physical object can result in a change in the environment by being either removed, added, moved, deformed or alter how it interacts with the sensor. However, this can be simplified to just Removed and Added. Any of the last three changes (moved, deformed, different sensor interaction) can be considered as a combination of removing the original item, and adding in the new. The distinction makes no difference from the robots point of view. Finally, an object can be observed to be exactly as expected - this will be denoted as Matched. There are many ways a sensor reading can be categorized as either of these, and the method chosen is by no means the only viable option.

B. The Environment Similarity Index (ESI)

The index is a ratio, based on the current pose estimate from the robots localization module - for this paper, AMCL. On every laser-scan callback, each ray point is evaluated: Is it close to an expected obstacle in the map? If yes, count as a Matched; else, count as an Added. This tells us that an object has been added to the environment that is not on the map. In addition, all laser rays that returned a value are traced in the map from the scanner and outwards to detect obstacles that they *should* have hit but didn't. These objects count towards Removed to indicate that something on the map has been removed from the environment. These classifcations are then used to calculate the ESI score; Figure X provides an overview of the classification.

C. Formal Definition

Matched: Let S be a 1D vector representing a single scan containing $n_s = |S|$ geometric scan points $s : S = (s_0, s_1, \ldots, s_{n_s-1})$. Only scan points that return a value within min and max range are considered valid. The points are indexed by i where $0 \leq i < n_s$. Let G be a 1D vector

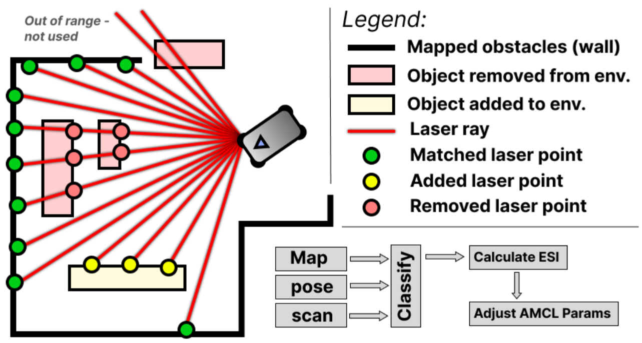

Fig. 1. Example of a figure caption.

representing the discrete geometric reference map, flattened from its original N-dimensional form:

$G = (g_0, g_1, \ldots, g_{n_g-1})$ The occupancy value at index i is $v(g_i)$, which equals 1 if g_i represents an obstacle, and 0 otherwise. The assignment $\lambda(i)$ returns the index in G with the shortest distance to the scan point s_i:

$$\lambda(i) = \underset{0 \leq j < n_g}{\arg\min}\ d(s_i, g_j) \tag{1}$$

The distance $d(p_1, p_2)$ is the Euclidean distance between two points in R^N, where N depends on the map representation, but any distance metric can be used. Let T be a variable distance threshold. It is based on the expected yaw noise γ [rad] and the range of the scan point $r(s_i)$:

$$T(i) = \sin(\gamma) * r(s_i) \tag{2}$$

Define a match as:

$$\chi_M(i) = \begin{cases} 1 & \text{if } d(s_i, g_{\lambda(i)}) < T(i), \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

The number of matches for a single scan is then:

$$M = \sum_{i=0}^{n_s-1} \chi_M(i) \tag{4}$$

Added: All valid scan points are classified as either Match or Added. Therefore, the number of added is simply the number of valid scan points minus the number of matches:

$$A = n_s - M \tag{5}$$

Removed: Each valid scan point defines the endpoint of a line from the scanner to that point, as seen in the world frame. The laser ray L contains all the discrete map points in G that lie on that line: $L = (l_0, l_1, \ldots, l_{n_r-1})$ where l_0 is closest to the scanner. We define a removed point to be every rising edge (going from free to occupied) that the ray encounters from the scanner and out. Define removed as:

$$\chi_R(j) = \begin{cases} 1 & \text{if } v(l_j) = 1 \text{ and } v(l_{j-1}) = 0 \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

The count of removed items (first encounters) in a scan with k rays is then:

$$R = \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j) \tag{7}$$

ESI: The ESI score for a single scan S is then defined as:

$$ESI \in [0, 1] = \frac{M}{M + A + R} \tag{8}$$

and since $n_s = M + A$, the final expression becomes:

$$ESI = \frac{\sum_{i=0}^{n_s-1} \chi_M(i)}{n_s + \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j)} \tag{9}$$

If the scan only sees matches, the ratio is $1/1 = 1$, indicating a perfect match with the map. Both added and removed points increases the denominator equally, driving the index down. Note how this reflects reality; there is no upper limit to how much you can change an environment; you can always use a sensor with a greater range and higher resolution and then add and remove more items. The index therefore converges towards zero as more changes are made, but it will only be exactly zero if it sees zero matches. The index can be calculated for any number of measurements per sensor update, and can be used for any given localization module. It is important to note that since the ESI depends on the current pose estimate, it will of course not be a correct representation of the environment if the pose error becomes too great. In order to confirm it's usability, we must show that the ESI accurately reflects the environment while the pose error is in its normal range. Then we need to show that ACML is able to maintain its pose error for no, light and moderate changes in the environment. We will confirm that the ESI can be calculated and used by a real robot in a real environment.

IV. Experiments

In order to test the method, two experiments are performed; one in a simulated environment, and one with a real robot in a real industrial environment. The experiments are designed to test the following:

- To what degree the ESI is a true measure of the true state of the environment.
- How strong is the correlation between ESI and the performance of AMCL.
- How can the ESI be used actively to reduce the pose error of AMCL.
- Whether the ESI is applicable and useful in the real world, and how it can be used to generate a detailed heatmap of local MED's.

A. Simulation experiment

The simulator used is Nvidia Isaac Sim v5.0. The robot stack runs on a separate docker container, and communicates with the simulator via ROS2 humble. The map is 50x50m, and contains 9 waypoints in a grid pattern. The direct pathways between waypoints are considered free space - obstacles are allowed to be placed in the remaining space, as seen in Figure X (red polygons). The experiment will consist of 30 individual runs. In each run, 20 boxes will spawn at random places with random orientations on the obstacle space. An occupancy grid map is then generated and used by the stack, which runs the default AMCL of Nav2 (ref). The robot will move randomly between waypoints, with random odometry noise added to the published odometry. While it moves, every fourth second, a randomly chosen box is moved (teleported) to a new random position in the obstacle space. This continues until all boxes have been moved. Then they are moved back into their original positions one by one, also every fourth second. The test ends when the last box is back in place. This recreates a scenario where the robot moves from a known to an unknown area, and then back again, meaning that in the middle of the test, the map-environment differences are at their highest, as none of the boxes are at their original positions. This is to capture not only when AMCL diverges, but also to see if it is able to recover. The robot has a single 360deg FoW 2D laser scanner with a max range of 30m. As such, it cannot cover the entire map from any one position. The "true" state of the environment is denoted Map Integrity Ratio (MIRA) and is calculated as

$$MIRA = \frac{U}{U + 2M} \tag{10}$$

where U is the number of unmoved boxes, and M is the number of moved boxes. M is scaled by 2 because a moved box accounts for the same change as an added plus a removed box. It is published as a ROS2 topic, and serves as a ground truth to evaluate how closely the ESI reflects the changes. Each run will be performed twice. One with the default AMCL parameters, and one where the ESI is used to dynamically tune the α-parameters of AMCL, which determine the magnitude of the noise added in the motion update step.

*1) **Hypothesis 1: ESI Validity and AMCL Performance**:* We hypothesize that:

1a) In the MIRA range of 1 to 0, the differences in the absolute pose error of AMCL is negligible for the majority of the range.

1b) Given a correct initial pose, and that the MIRA remains moderate or below, the ESI will be able to accurately reflect the true level of divergence. First we plot the relationship between the MIRA, ESI and the pose and yaw errors of AMCL for 100% of the data to understand the patterns. Then we will load the first 50% of all runs to ensure that the pose errors for an entry is unaffected by any previous time instance where the MIRA and ESI score were lower. To test 1a, we sort all entries into 10 bins based on their MIRA value. Then we plot the mean pose error for each group, see figure X, and evaluate the findings. To test 1b, we will perform a Spearman's rank correlation test on the ESI and the MIRA to determine if there is a strong correlation.

Results: The ESI scores for the individual runs and their interpolated mean (black) are shown in Figure X (top), along with the ground truth MIRA value (red). The absolute position and yaw errors are also shown (bottom left and right). In general, the scores follow the MIRA closely in the first half of the test. The general pattern and stepwise changes is clearly reflected in the mean, with no visible time delay. In the beginning, since the ESI can never be more than 1, the mean is slighly below the true value. It then follows it very closely, and after about 15 seconds, the ESI rises slightly above the MIRA, which persists until just before half way through the test. Since the robot cannot observe the entire map all the time, this difference is expected: On average, there will be changes outside the range of the laser, which means that, since the environment state is only declining, the true state of the environment can only be as bad or worse than what the robot can detect. When MIRA reaches zero, the ESI is still above it, even though the map and environment are now completely different. This is caused by false positive matches and missed removed objects that results from the rising pose error, and because when a box is moved, it may randomly be placed such that it overlaps with the initial

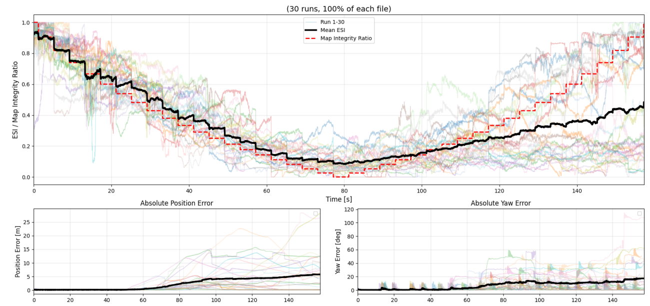

Fig. 2. Top: ESI vs MIRA for default AMCL. Red = Ground truth (MIRA). Bottom left: Position error. Bottom right: Yaw error. Black = interpolated mean of the ESI, position error and yaw error, respectively.

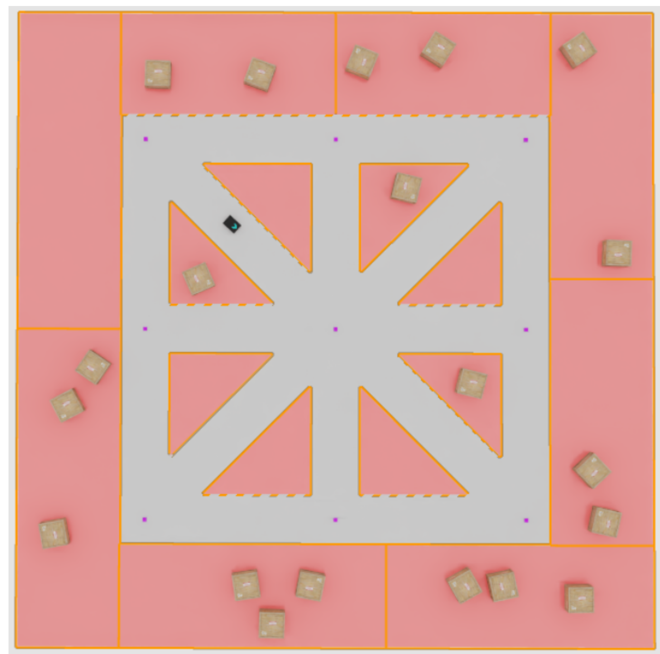

Fig. 3. Scene from Isaac Sim. Red polygons represent the obstacle space. Waypoints are marked with purple squares. The robot is seen moving from top left to center.

(mapped) position of another box, thus contributing to a match, also as a false positive. After this, the pose errors becomes too great, and the ESI can no longer be trusted.

Up until the 50s mark, both position and yaw error is kept low, with mean values of 0.1m and 0.57°, respectively. Yaw does exhibit some spikes to around 10 degrees, but they are quickly corrected. At this point, the ESI is about 0.27 (MIRA = 0.21). After this, they both start to increase rapidly.

1a Anova results
TODO, but clearly negligible until the 0.3-0.4 bin.

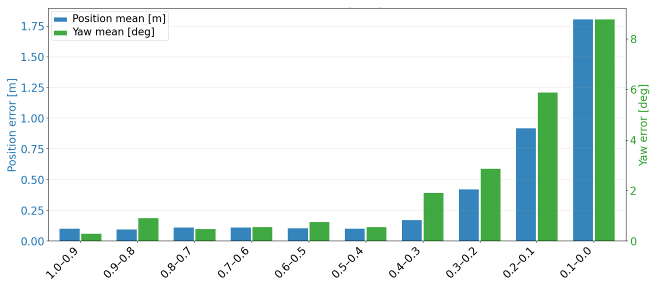

Fig. 4. Balanced (N=12602 in each bin) bar plot of mean position and yaw error for each MIRA bin.

1b Spearman's rank correlation test results
TODO, but results:
rho : 0.99838
p : 0.000
$Number\ of\ pairs$: 4718
This shows that, *as long as the map and environment are at least 40% similar, the default AMCL is able to provide accurate pose estimates in changing environments.* It also shows that, *given the robot is well localized, as long as the ESI is kept above 0.3, we can expect it to be an accurate estimate of the map-environment difference.* However, there will be scenarios where AMCL has diverged, even with high ESI scores, due to other factors such as map symmetry, lack of features, hallway areas, etc.

*2) **Hypothesis 2: Correlation between ESI and pose error**:* We hypothessize that there is a strong correlation between the observed ESI and the pose RMSE of default AMCL. To test this, we sample the first 60% (90 seconds) of all 30 runs. The remaining 40% is ommited, because as observed, from this point on, the ESI can no longer be trusted to accurately reflect the MIRA, due to the high pose error. The runs are then time synchronized, and for each time step, we generate a sample point with two values: The mean ESI and the mean pose RMSE for that time across all 30 runs. This results in 5662 pairs of data points, which can be seen in Figure X. As observed in both figure X (this) and Y (esi vs MED), we do not expect a linear correlation, and the data is not normally distributed. Therefore, we perform a Spearman's rank correlation test, with null-hypothesis significance level $\alpha = 0.001$.

Results: The correlation coefficient is $\rho = -0.83998$ with a p-value of $p = 0.00000$ (arithmetic underflow). This shows that there is a very strong negative correlation between the

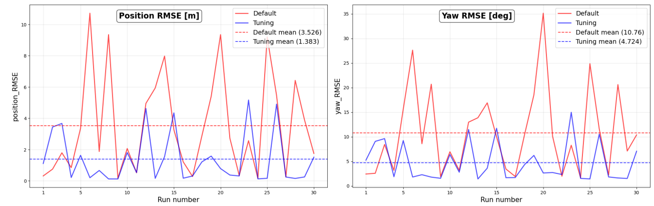

Fig. 5. Alpha parameter tuning results.

mean ESI score and pose RMSE across all runs. (footnote to raw data?)

*3) **Hypothesis 3: ESI for tuning AMCL parameters**:* The hypothesis is that the α-parameters of AMCL that controls the noise added to the motion update step can be dynamically adjusted live to reduce the pose error that would otherwise increase in rapidly changing environments. The reasoning is that when the environment does not match the map, the particle cloud will continue to expand, because none of them have a significantly higher weight than the others, and therefore the filter will not converge. But if some of the particles on the edge of the cloud, where the accumulated random noise is highest, suddenly have just a few beams with a high weight (false positives), the filter will quickly converge to that wrong pose, and then be unable to recover. By using the ESI to limit this expansion, this risk is reduced. Essentially, AMCL now places more trust in the odometry, because the ESI indicates that the laser data can no longer be trusted. When (if) the environment slowly begins to match the map again, the ESI will rise, returning the filter to its default level of exploration. The two runs are analyzed and compared.

Results The results can be seen in figure X. The results shows that tuning the parameters reduced the mean position and yaw RMSE by 60.77% and 56.09%, respectively. Although it does not completely prevent divergence, it is still a significant improvement that may in some cases be a decisive factor.

B. Real-world experiment

To demonstrate that the ESI can be applied in the real world, and that it can be used to visualize changes in it, we used a MiR100 AMR as the platform. It has two SICK S300 safety rated 2D Lidar sensors that scans in a horizontal plane 20cm above the ground, with a 360 degree field of view and a max range of 30m. The venue is the production hall of a packaging and labeling company. It was chosen because it represents the type of environment where AMCL often struggles; it is constantly developing, and most of the time, the the majority of what the robot is able to see consists of transient objects, as seen in Figure X. A small section of the hall was mapped, using the native mapping functionality (based on ROS1 cartographer). Two waypoints, Start and Finish, were placed. The task of the robot was simply to move from Start to Finish. In order avoid disrupting their production, instead of moving the items around, we first recorded the map, and then gradually edited it manually, resulting in similar map-environment disagreements. Some transient objects were removed, and new ones added. The robot was tested in four scenarios: 1) No, 2) light, 3) moderate and 4) severe changes. Each scenario was executed 5 times, and the ESI recorded for all runs. When the robot reached the end, its pose error was obtained from a reference coordinate system on the floor. The ESI scores found during each run were then used to create heatmaps for each scenario, as seen in figure X. The method used is Inverse Distance Weighting (IDW) with a power of 2 and radius of 30m. A Line-of-sight (LoS) check was implemented in order to reflect how readings on either side of an obstacle should not affect each other.

*1) **Hypothesis 4: Real-world usefulness and heatmap visualization**:* hyp and setup text

TABLE I

ESI RESULTS FOR THE REAL-WORLD EXPERIMENT. 1) NO CHANGES, 2) LIGHT CHANGES, 3) MODERATE CHANGES, 4) SIGNIFICANT CHANGES. ROWS MARKED WITH * HAVE UNITS [mm] AND [°], RESPECTIVELY.

Scenario	1)		2)		3)		4)	
* Pose RMSE	22.4	2.4	17.5	2.1	21.1	2.4	N/A	N/A
* Pose Std. dev.	25.1	2.7	19.5	1.1	20.5	1.0	N/A	N/A
ESI mean	0.94		0.64		0.46		0.22	
ESI std. dev.	0.04		0.12		0.10		0.09	

Fig. 6. Photo of a MiR100 moving through the production hall towards the finish waypoint.

*2) **Results**:* The test data reveals that a large percentage of the emitted rays are never returned, and so they report max range. This is why only those beams that return a value within min and max range are considered valid: A raytraced beam that reports max range due to attenuation would wrongly count all objects on the line as removed.

To test the reliability of AMCL, a Levene's test of equality of variances on the position errors (Euclidean distance from goal) for scenarios 1-3 shows no significant differences in variance (p = 0.8153, center=median). A one-way ANOVA test shows no significant differences in mean position error (p = 0.7046). These tests show that, in this experiment, for no, light and moderate changes, AMCL is able to consistently maintain a low pose error. However, when the changes became severe, AMCL failed in every run, and the robot was unable to reach the goal. This corresponded with a low mean ESI score of 0.22 for scenario 4.

The heatmap clearly shows how the ESI reflects the intensity of local changes across the hall in the different scenarios.

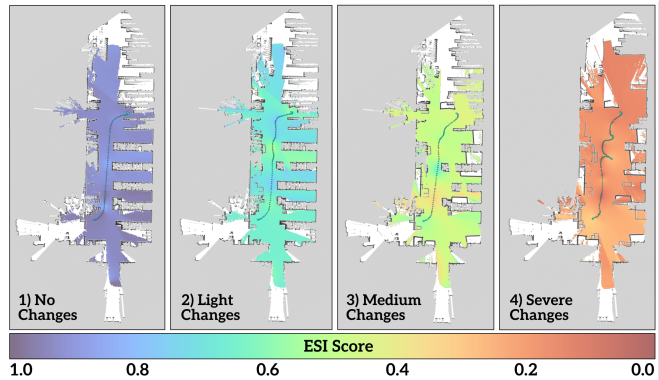

Fig. 7. ESI-based Heatmaps for each of the four scenarios in the same section of the hall.

V. CONCLUSIONS

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VI. LIMITATIONS AND FUTURE WORK

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APPENDIX

- Raw position data for the real-world experiment - Mean ESI and mean RSME data for Spearman's rank correlation test

ACKNOWLEDGMENT

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